

2(a)	(resultant) force proportional/equal to/is rate of change of momentum	B1
2(b)(i)	distance = area under graph or $s = \frac{1}{2} (u + v) t$ $= \frac{1}{2} \times (9 + 13) \times 10$ or $s = ut + \frac{1}{2}at^2$ $= (9 \times 10) + (\frac{1}{2} \times 0.40 \times 10^2)$ or $s = vt - \frac{1}{2}at^2$ $= (13 \times 10) - (\frac{1}{2} \times 0.40 \times 10^2)$ or $v^2 = u^2 + 2as$ $13^2 = 9^2 + (2 \times 0.40 \times s)$	C1
	distance = 110 m	A1

Question	Answer	Marks
2(b)(ii)	1. $a = \text{gradient}$ or $a = (v - u) / t$ or $a = \Delta v / (\Delta)t$ e.g. $a = (14 - 9) / 12.5$ or $(13 - 9) / 10$	C1
	$a = 0.40 \text{ m s}^{-2}$	A1
	2. resultant force $= 850 \times 0.40$ $= 340 \text{ N}$	A1
	3. $(F =) 510 + 440 + 340 = 1300 \text{ (N)}$	A1
	4. $P = Fv$	C1
	$= 1300 \times 13$ $= 1.7 \times 10^4 \text{ W}$	A1
2(c)	$E = \sigma / \epsilon$	C1
	$E = (F / A) / (\Delta L / L)$ or $E = FL / A\Delta L$	C1
	$\Delta L = (480 \times 0.48) / (3.0 \times 10^{-4} \times 2.2 \times 10^{11})$ $= 3.5 \times 10^{-6} \text{ m}$	A1
2(d)	$f_o = f_s v / (v - v_s)$ $480 = f_s \times 340 / (340 - 14)$	C1
	$f_s = 460 \text{ Hz}$	A1